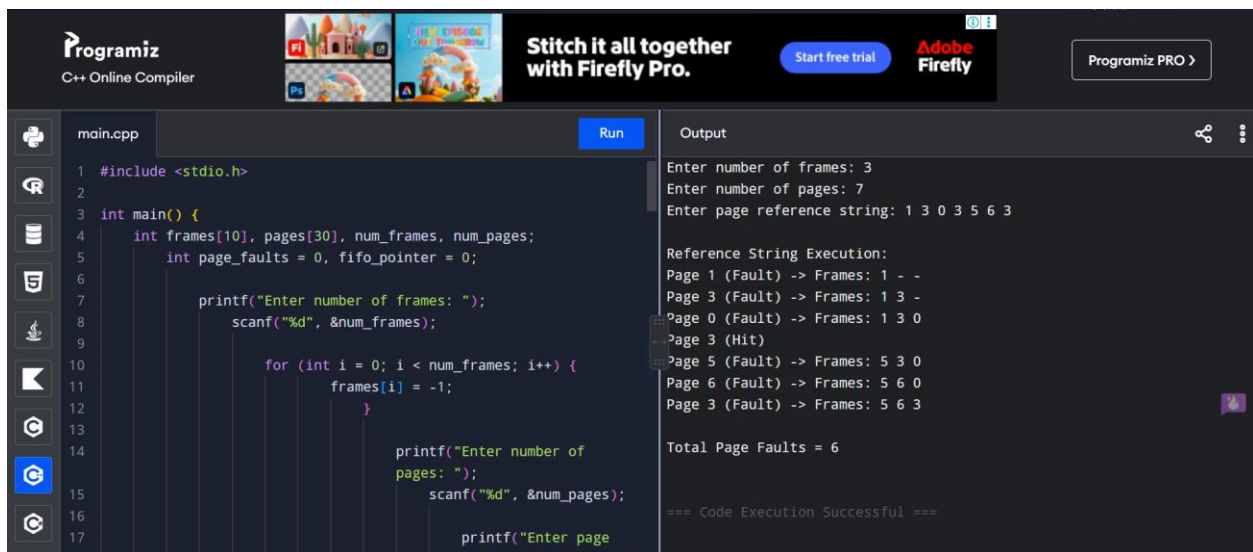


EXPERIMENT 25:

25. Construct a C program to simulate the First in First Out paging technique of memory management.

AIM:

To construct a C program to simulate the First In First Out (FIFO) page replacement technique of memory management.



The screenshot shows the Programiz C++ Online Compiler interface. The code editor on the left contains a C program for simulating the FIFO page replacement technique. The output window on the right shows the execution results.

```
1 #include <stdio.h>
2
3 int main() {
4     int frames[10], pages[30], num_frames, num_pages;
5     int page_faults = 0, fifo_pointer = 0;
6
7     printf("Enter number of frames: ");
8     scanf("%d", &num_frames);
9
10    for (int i = 0; i < num_frames; i++) {
11        frames[i] = -1;
12    }
13
14    printf("Enter number of
15    pages: ");
16    scanf("%d", &num_pages);
17
18    printf("Enter page
19    reference string: ");
```

Output:

```
Enter number of frames: 3
Enter number of pages: 7
Enter page reference string: 1 3 0 3 5 6 3

Reference String Execution:
Page 1 (Fault) -> Frames: 1 - -
Page 3 (Fault) -> Frames: 1 3 -
Page 0 (Fault) -> Frames: 1 3 0
Page 3 (Hit)
Page 5 (Fault) -> Frames: 5 3 0
Page 6 (Fault) -> Frames: 5 6 0
Page 3 (Fault) -> Frames: 5 6 3

Total Page Faults = 6

=== Code Execution Successful ===
```

PSEUDO CODE :

BEGIN

PRINT "Enter number of frames:"

READ num_frames

DECLARE frames array of size num_frames

INITIALIZE frames with -1 (indicating empty)

PRINT "Enter number of pages:"

READ num_pages

DECLARE pages array of size num_pages

```

    PRINT "Enter page reference string:"
FOR i FROM 0 TO num_pages - 1 DO
    READ pages[i]
END FOR

SET page_faults = 0
SET fifo_pointer = 0

FOR i FROM 0 TO num_pages - 1 DO
    SET page_found = 0
    SET page = pages[i]

    // Check if page already exists in frame
    FOR j FROM 0 TO num_frames - 1 DO
        IF frames[j] == page THEN
            SET page_found = 1
            BREAK
        END IF
    END FOR

    // If page is not in frame (Page Fault)
    IF page_found == 0 THEN
        SET frames[fifo_pointer] = page
        SET fifo_pointer = (fifo_pointer + 1) MOD num_frames
        INCREMENT page_faults
    END IF

    PRINT "Page " + page + " -> Frames: " + CURRENT_FRAMES_STATE
ELSE
    PRINT "Page " + page + " -> Page Hit"

```

```
END IF  
END FOR
```

```
PRINT "Total Page Faults: " + page_faults  
END
```

CODE:

```
#include <stdio.h>
```

```
int main() {
```

```
    int frames[10],  
    pages[30], num_frames,  
    num_pages;
```

```
    int page_faults = 0,  
    fifo_pointer = 0;
```

```
    printf("Enter number  
of frames: ");
```

```
    scanf("%d",  
&num_frames);
```

```
    for (int i = 0; i <
num_frames; i++) {

        frames[i] = -1;

    }
```

```
    printf("Enter number
of pages: ");
```

```
    scanf("%d",
&num_pages);
```

```
    printf("Enter page
reference string: ");
```

```
    for (int i = 0; i <
num_pages; i++) {

        scanf("%d",
&pages[i]);

    }
```

```
    printf("\nReference
```

```
String Execution:\n");
```

```
    for (int i = 0; i <  
num_pages; i++) {
```

```
        int page_found = 0;
```

```
        for (int j = 0; j <  
num_frames; j++) {
```

```
            if (frames[j] ==  
pages[i]) {
```

```
                page_found =  
1;
```

```
                break;
```

```
            }
```

```
        }
```

```
        if (page_found ==  
0) {
```

```
            frames[fifo_pointer] =  
pages[i];
```

```
        fifo_pointer =  
(fifo_pointer + 1) %  
num_frames;  
  
        page_faults++;  
  
        printf("Page %d  
(Fault) -> Frames: ",  
pages[i]);  
  
        for (int k = 0; k  
< num_frames; k++) {  
  
            if (frames[k]  
!= -1)  
  
                printf("%d  
", frames[k]);  
  
            else  
  
                printf("- ");  
  
        }  
  
        printf("\n");  
  
    } else {
```

```
        printf("Page %d  
(Hit)\n", pages[i]);  
  
    }  
  
}  
  
printf("\nTotal Page  
Faults = %d\n",  
page_faults);  
  
return 0;  
  
}
```